What Position Makes or Breaks an NBA Team?

Conventional wisdom says you build around a big or a wing. The on/off data says the *downside* is the same everywhere — but the *upside* lives at point guard, and the numbers quietly lie about centers.

METHOD · RAPTOR on/off ± regularized blend SOURCES · FiveThirtyEight · Basketball-Reference

The question sounds simple: if you could only get one position right, which one swings your season the most? "On/off" splits are the natural tool — they measure how a team's net rating changes with a player on the floor versus on the bench. But "makes or breaks" hides two different questions, and the data answers them differently.

The first is **leverage**: does a slot matter more on average? The second is **swing**: is the gap between a great and a poor player at that slot the widest — so that getting it right versus wrong moves the needle most? Across 1109 rotation player-seasons, here is what separated the positions — and what didn't.

+8.0
Mean on/off net of the **top-10% point guards** — the highest ceiling of any position (next best: +6.8).

8.9–9.7
Points-per-100 **make-or-break swing** (P10 → P90) — and it's nearly identical at *every* position. The downside isn't slot-specific.

-0.29 → +0.13
Center's mean impact, **raw on/off → regularized**. The biggest correction of any position: on/off undersells bigs.

01 The downside is position-agnostic

DISTRIBUTION OF ON/OFF NET IMPACT, BY POSITION

Plot every rotation player's on/off net and the five distributions look almost interchangeable. The P10–P90 swing runs from **8.9** to **9.7** points per 100 possessions — a spread so tight that a formal test for unequal variance across positions comes back **non-significant** (Levene `p = 0.5238`). Differences in the *average* are non-significant too (ANOVA `p = 0.2663`; Kruskal–Wallis `p = 0.3791`).

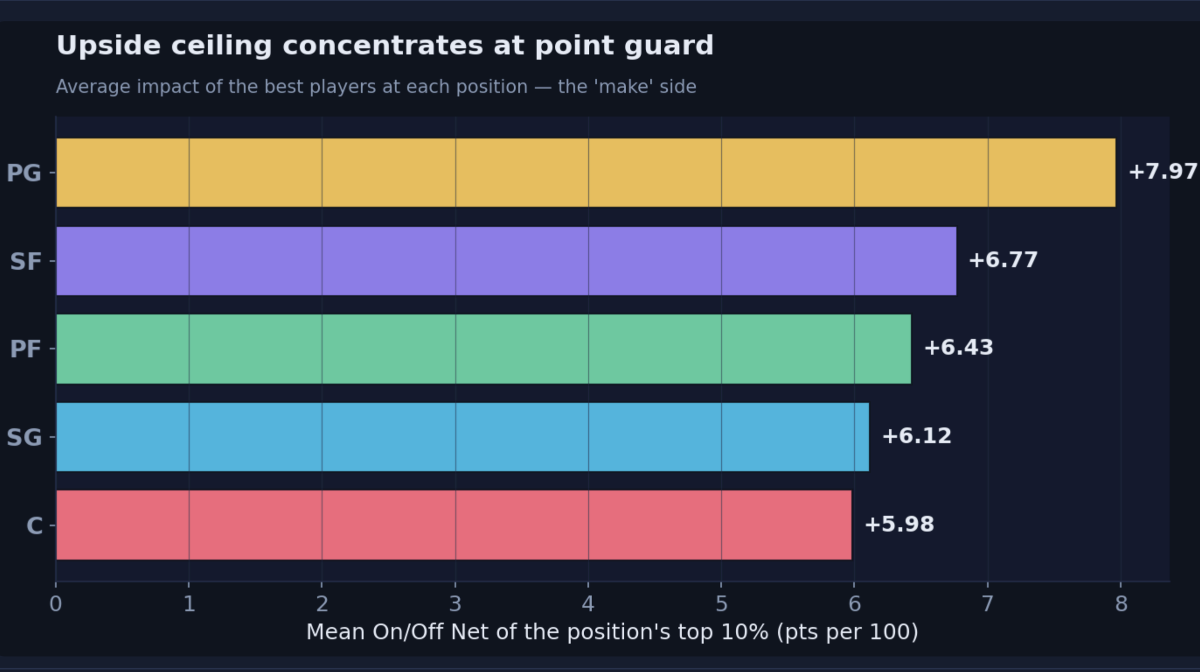
Each violin is one position's full on/off distribution. Bars span P10 → P90; ● marks the median, ▲ the P95 ceiling. Note the lone exception to the symmetry: point guard's upper tail runs higher than anyone's.

> The blunt read: at the rotation level, **fielding a bad starter costs you roughly the same ~9 points per 100 no matter which position he plays**. There is no single slot where a weak link is uniquely fatal.

02 But the ceiling belongs to point guards

MEAN IMPACT OF EACH POSITION'S TOP 10% — THE "MAKE" SIDE

Symmetry breaks at the top. Isolate the best 10% at each position and the point guards pull away: their elite tier averages **+8.0** on/off net, against +6.8 for the next-closest group and roughly +6 for everyone else. The highest individual seasons in the sample — Stephen Curry, Chris Paul — are lead guards, and crucially they stay elite even after the noise is regularized away.



The average of the very best at each slot. Point guard's edge here — not in the middle of the distribution — is where "build around a floor general" earns its keep.

Point Guard	Shooting Guard	Small Forward	Power Forward	Center
Stephen Curry 2015 +12.3 reg +11.0	Kyle Korver 2015 +9.3 reg +4.2	LeBron James 2017 +13.5 reg +7.8	Draymond Green 2016 +15.2 reg +9.4	DeAndre Jordan 2015 +8.9 reg +4.4
Stephen Curry 2017 +12.2 reg +9.2	Manu Ginobili 2014 +7.4 reg +5.9	LeBron James 2016 +12.7 reg +8.2	Draymond Green 2015 +10.3 reg +7.3	DeAndre Jordan 2017 +8.8 reg +2.8
Chris Paul 2015 +11.4 reg +10.7	Bradley Beal 2017 +7.3 reg +5.1	Jae Crowder 2014 +9.5 reg +3.1	Draymond Green 2017 +8.4 reg +7.8	Nikola Jokic 2017 +7.7 reg +7.3

Each cell: the player's raw on/off net, with the regularized RAPTOR figure beneath. Watch how some marks shrink hard once regularized (Kyle Korver +9.3 → +4.2) while others barely move (Curry +12.3 → +11.0) — a quick visual test of who's real.

03 The make-or-break swing, ranked

REPLACEMENT LEVEL (P10) → ELITE (P90), PER POSITION

The full swing from a 10th-percentile to a 90th-percentile player at each slot. The ordering is real but the magnitudes are close — consistent with the statistical tests finding no significant variance gap.

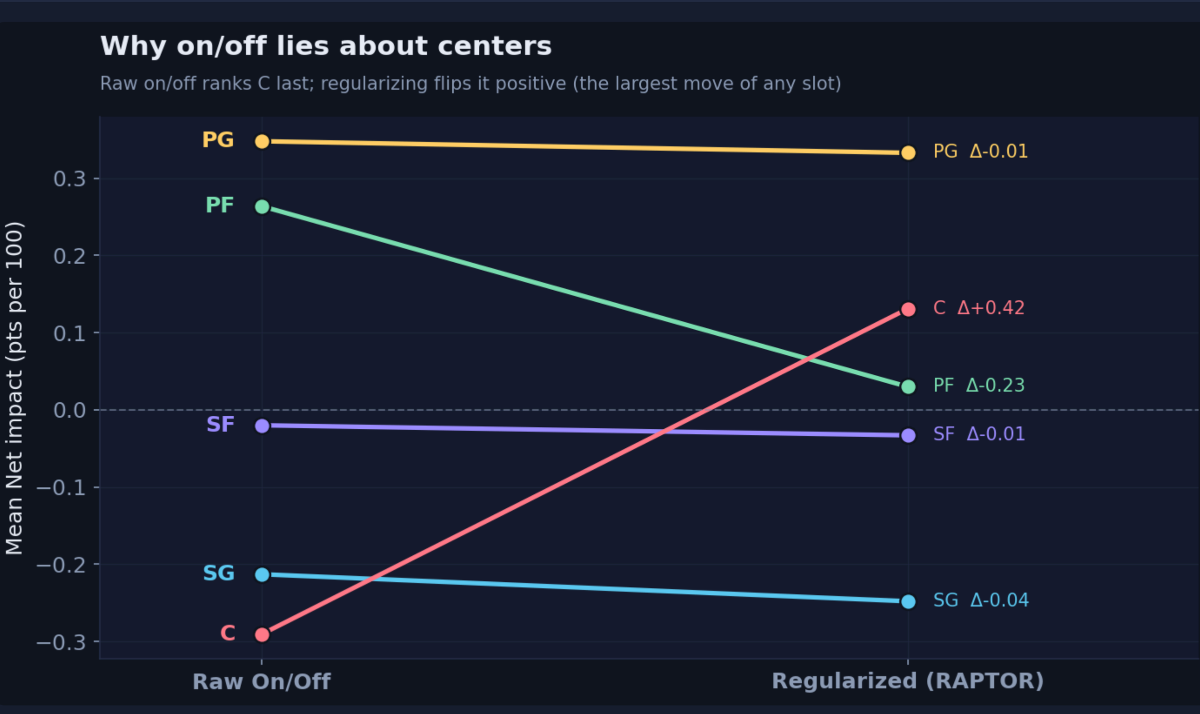
POS	N	MEAN	SD	P10	P90	SWING	P95 CEILING	REG. MEAN
PG	233	+0.35	3.87	-3.94	+5.01	8.95	+7.58	+0.33
SG	242	-0.21	3.71	-4.95	+4.79	9.75	+6.10	-0.25
SF	220	-0.02	3.56	-4.33	+4.53	8.86	+5.74	-0.03
PF	212	+0.26	3.62	-4.53	+4.44	8.97	+5.64	+0.03
C	202	-0.29	3.61	-5.00	+4.04	9.05	+5.78	+0.13

On/off net in points per 100 possessions. **Swing** = P90 – P10. **Reg. mean** is the regularized RAPTOR blend, shown as a noise check on the raw on/off column.

04 Why on/off lies about centers

RAW ON/OFF VS. THE REGULARIZED BLEND

Here is the trap. Read raw on/off alone and **center grades out worst of the five positions** (mean -0.29). Regularize the same data — adjusting for who's on the floor around each player — and center flips to **+0.13**, the single largest move of any position (`Δ +0.42`). Power forward shifts the same direction.



Each line is one position's mean impact under the two methods. Center (pink) crosses from bottom to positive; the guards and wings barely move. When a metric reorders your conclusion, the metric — not the player — is the story.

> This is the methodological heart of the project. Raw on/off rewards rim-running, elite-rebounding bigs partly for *who replaces them* and for lineup collinearity — DeAndre Jordan's +8.9 raw season regularizes to +4.4. It's exactly why impact sites turn raw on/off into RAPM before trusting it. **Take on/off at face value and you'd crown the wrong position.**

05 Offense is the lever — especially at the point

WHERE EACH POSITION'S VALUE SWINGS: OFFENSE VS. DEFENSE

Splitting the swing into its offensive and defensive halves, every position derives slightly more of its variance from offense — but the tilt is steepest for point guards (offense 59% of swing). The interior positions split more evenly, consistent with bigs banking value on the defensive end that on/off captures only partially.

P10–P90 swing decomposed. The takeaway pairs with §2: the point guard ceiling is built on offensive creation, the hardest skill to replace.

The verdict

So, which position makes or breaks a team? The honest answer is a split decision. **No position carries more downside risk than the others** — a replacement-level starter is about equally damaging anywhere. What separates the slots is the **ceiling**, and that ceiling is highest at **point guard**: the best lead guards swing a team further, and more durably under scrutiny, than the best players at any other position. Meanwhile the position on/off *seems* to break a team — center — is mostly a measurement artifact.

Put plainly: you don't lose a season by getting any one position wrong more than another, but you *win* one most reliably by getting the point guard right — and you should never trust a raw on/off number about a big man without regularizing it first.

REPRODUCIBLE

Data & method

Population: every NBA player-season 2014–2017 appearing in 538's RAPTOR file, joined to Basketball-Reference season positions by normalized name. Filtered to `MP ≥ 1000` (rotation players), leaving **1109** player-seasons. Positions taken per-season (primary listed position for hybrids), grouped into the five classic slots.

Metrics

On/Off net = `raptor_onoff_total` (team net-rating change, on minus off, points per 100 possessions), with offense/defense components. **Regularized blend** = `raptor_total`, RAPTOR's noise-adjusted estimate, used throughout as a cross-check so conclusions don't ride on raw-on/off noise. **Swing** = P90 – P10 within a position.

Tests

Equality of means across positions: one-way ANOVA (`p=0.2663`) and Kruskal–Wallis (`p=0.3791`). Equality of variance — the make-or-break claim: Levene's test on medians (`p=0.5238`). Robustness re-run at `MP ≥ 1500` preserved the ordering.

Sources

Impact / on-off: FiveThirtyEight RAPTOR. Positions: Basketball-Reference season tables (`Seasons_Stats.csv`). Conceptual framing & the regularize-your-on/off principle follow the public methodology at nbarapm.com and databallr.

Caveats

On/off is confounded by teammate and replacement quality; a four-season window is a snapshot, not a law; and positional labels are increasingly fuzzy in the modern NBA. The regularized cross-check mitigates the first; the consistent direction of findings under the stricter minutes cut mitigates the second.